

9. (a) Ethanedioic acid (oxalic acid) has the formula $(\text{COOH})_2$.

It can be made in the laboratory by oxidising sugar with a powerful oxidising agent (shown as [O] in the equation below).

- (i) Balance the equation. [1]



- (ii) After purification, the oxalic acid is produced as white crystals of a hydrate, $(\text{COOH})_2 \cdot x\text{H}_2\text{O}$. On heating this hydrate, water is lost, leaving anhydrous oxalic acid.



In an experiment 5.67 g of the hydrate were heated to constant mass, giving 4.05 g of the anhydrous acid.

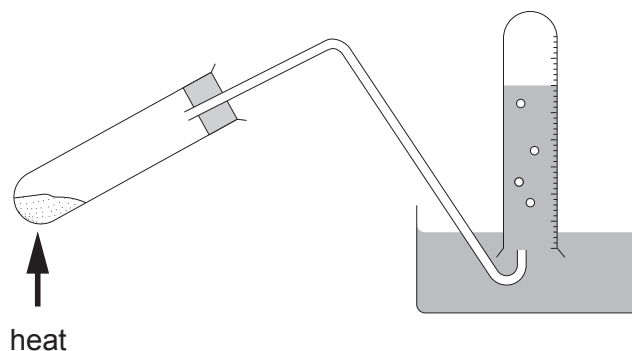
Calculate the value of x in $(\text{COOH})_2 \cdot x\text{H}_2\text{O}$. You **must** show your working. [3]

$x = \dots\dots\dots$

- (b) Another method of producing oxalic acid in the laboratory is to heat potassium methanoate, HCOOK . This produces potassium oxalate, which is then acidified to give the acid.



Some students carried out this experiment using the apparatus below. They collected the hydrogen gas produced over water.



A few students obtained a smaller volume of hydrogen than expected.

Suggest **two** practical reasons why this may have occurred, apart from leaks. [2]

1

2

(c) Gareth was asked to analyse a mixture of potassium methanoate and potassium oxalate. He weighed out 4.69 g of the mixture and carried out the method below.

- The mixture was completely dissolved in distilled water.
- Calcium chloride solution was added to this solution (only calcium oxalate was precipitated).
- The mixture was filtered and the precipitated calcium oxalate washed and dried.

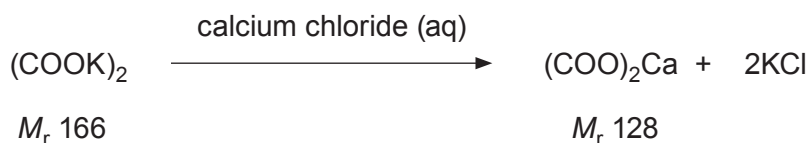
(i) Suggest how Gareth would know when enough calcium chloride solution had been added to react with all of the potassium oxalate present. [2]

.....

.....

.....

(ii) In his experiment Gareth obtained 2.49 g of pure dry calcium oxalate (M_r 128).



Calculate the percentage of potassium oxalate in the mixture, giving your answer to the appropriate number of significant figures. [3]

Percentage of potassium oxalate = %

